U^{2}DP: Unlocking Unlabeled Data Potential for Semi-Supervised Remote Sensing Image Captioning

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***Abstract*—Remote sensing image captioning (RSIC) aims to generate concise and accurate textual descriptions for complex aerial scenes, yet faces a critical challenge of limited labeled training data. To address this data scarcity, a semi-supervised RSIC that leverages unlabeled data (i.e., unlabeled images) has attracted significant attention. However, existing semi-supervised methods often rely on sample filtering strategies for handling unlabeled data, potentially discarding valuable information from unselected samples. To address this limitation, we propose U^{2}DP, a novel semi-supervised RSIC framework that comprehensively Unlocks Unlabeled Data Potential via adaptive supervision reconstruction. Specifically, we first leverage the pretrained CLIP model to evaluate the expressive capability of the caption generator on unlabeled data, categorizing them into three levels, from fine-grained to coarse-grained: strongly matched, weakly matched, and unmatched. Via this evaluation, we construct different pseudosupervision signals for unsupervised learning, including the model's own predictions and the pseudolabel predicted by CLIP from a remote sensing vocabulary. Therefore, U^{2}DP enables effective learning from unlabeled data without sample filtering, further improving model performance. Extensive experiments are conducted on three widely-used public RSIC benchmark datasets, including Sydney-Captions, UCM-Captions, and RSICD. The results consistently demonstrate that our U^{2}DP is superior to existing methods across all datasets. The source code is publicly available at https://github.com/njustkmg/U2DP**

***Index Terms*—Cross-modal learning, remote sensing image captioning (RSIC), semi-supervised learning (SSL).**

I. Introduction

REMOTE sensing image captioning (RSIC) has emerged as a pivotal task in the remote sensing domain, aiming to generate concise and meaningful textual descriptions of complex scenes captured in satellite and aerial imagery [1]. By integrating advancements in computer vision and natural language processing, RSIC bridges the gap between visual perception and linguistic expression, enabling automated interpretation of remote sensing data. Unlike traditional remote sensing tasks, such as scene classification [2], [3], object detection [4], [5], and target segmentation [6], [7], which primarily focus on recognizing specific objects or categorizing scenes, RSIC provides a more comprehensive understanding of aerial imagery through natural language descriptions. This capability is particularly valuable for applications such as land-use classification, disaster assessment, and forestry resource mapping, where textual descriptions enhance interpretability and facilitate decision-making.

Despite its potential, RSIC presents challenges that differ substantially from those in general-purpose image captioning [8], [9]. First, remote sensing imagery often contains dense, multiscale scenes with numerous small objects, making it difficult to capture all salient details. Second, the task is complicated by high interclass similarity and large intraclass variation, where different land-use categories may appear visually alike, while the same category can vary greatly across regions, seasons, or imaging conditions. Third, RSIC requires domain-specific semantic distinctions, such as differentiating between a "dockyard" and a "container port," which are visually similar but functionally distinct. These unique factors make caption generation particularly challenging and heavily dependent on expert interpretation for reliable manual annotation. Consequently, large-scale remote sensing image–text pairs are still scarce, while vast archives of unlabeled imagery remain largely underutilized.

This imbalance between scarce paired data and abundant unpaired imagery naturally motivates the adoption of semi-supervised learning (SSL) for RSIC. The central challenge is to design a robust framework that can extract reliable and informative supervision signals from automatically generated captions for unlabeled images, enabling them to effectively complement limited labeled data. However, the unique characteristics of RSIC make the direct transfer of general SSL methods suboptimal. Many existing techniques [10], [11], [12] that perform well on natural images struggle in RSIC because they fail to leverage partially correct yet noisy captions, and they lack robust, domain-aware fallback mechanisms when captions are entirely unreliable.

Current SSL strategies for image captioning can be broadly divided into two categories.

1) Pseudolabeling methods, such as the noisy student framework [13], retain only high-confidence predictions

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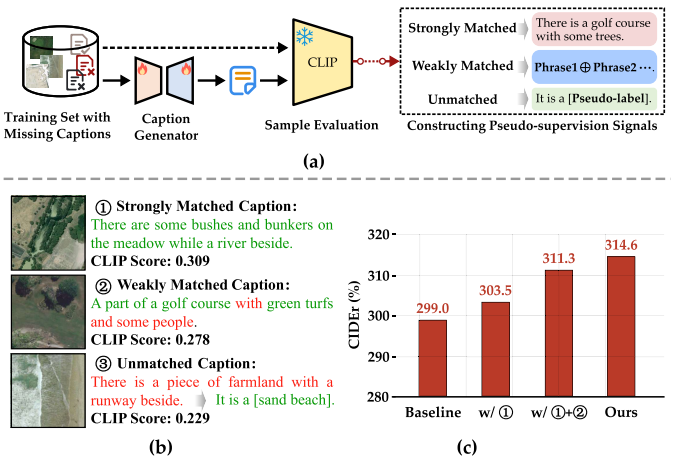


Fig. 1. (a) Illustration of SSL on unlabeled data. We first evaluate the quality of the generated captions via CLIP model, and then construct corresponding pseudosupervision signals. (b) Tri-level caption quality: strongly matched, weakly matched, and unmatched. Words highlighted in green are treated as different pseudosupervision signals, while those in red represent information that is discarded. (c) Performance gains in SSL with captions of varying quality levels on the UCM-Captions dataset. Note that "Baseline" and "Ours" represent the cases where no unlabeled data and all unlabeled data are used, respectively.

while discarding those with lower confidence. In RSIC, however, such low-confidence predictions often arise from dense- or fine-grained scenes that—despite lower confidence—still contain valuable object-level or relational cues. Captioning-specific adaptations, like object dropout with paraphrasing [14], refine pseudolabels to improve quality, yet they remain constrained by the same limitation of discarding potentially useful low-confidence outputs.

2) Consistency-based methods, such as mean teacher [15], enforce prediction stability under small input perturbations. In RSIC, though, partially incorrect captions or substantial domain shifts can cause this regularization to inadvertently reinforce errors. Even advanced variants like cross-modal relation consistency (CPRC) [11], which aligns images and captions in a shared semantic space, still lack mechanisms to adaptively handle varying pseudolabel quality and to incorporate reliable domain-specific cues when needed.

To address these issues, we propose U^{2}DP, a semi-supervised RSIC framework designed to fully exploit unlabeled data through a hierarchical pseudosupervision mechanism guided by image–text alignment confidence. Specifically, as shown in Fig. 1(a), we integrate a pretrained vision–language CLIP model [16] to compute an alignment score, implemented as the semantic similarity between each generated caption and its corresponding image. Via this score, captions are categorized into three quality levels: strongly matched, weakly matched, and unmatched [see Fig. 1(b)]. Strong matches are directly adopted in full as pseudolabels; weak matches are distilled into their most relevant object and relational phrases, preserving informative content while suppressing noise; and unmatched cases are replaced by minimal, low-noise prompts constructed from a domain-specific remote sensing vocabulary (e.g., `[It is a __.]`), ensuring that even in the absence of a reliable caption, the model benefits from meaningful categorical cues without introducing misleading details. This tri-level pseudo-supervision strategy dynamically adjusts learning based on caption quality, enabling the model to extract useful signals from imperfect captions and avoid error accumulation from noisy pseudolabels. Therefore, U^{2}DP not only alleviates reliance on scarce image–text pairs but also enhances the caption generator's adaptability to varying data quality conditions, ultimately improving RSIC performance [see Fig. 1(c)]. To sum up, our contributions are threefold.

1) We propose U^{2}DP, a universal semi-supervised RSIC framework that fully exploits unlabeled data and can be integrated with any captioning model.
2) We develop a CLIP-based mechanism to evaluate image–text alignment and reconstruct tailored pseudo-supervision signals, facilitating effective unsupervised learning.
3) Extensive experiments on three RSIC datasets demonstrate consistent performance improvements, validating the framework's effectiveness in scenarios with limited labeled data.

II. Related Work

A. Remote Sensing Image Captioning

RSIC is one of the most active research topics in remote sensing vision–language research. Existing approaches for RSIC can be broadly divided into three categories: template-based methods [17], retrieval-based methods [18], and encoder–decoder-based methods [19], [20], [21], [22].

Template-based methods treat caption generation as an object detection and slot-filling problem. A predefined set of sentence templates with placeholders is first constructed, and fully convolutional networks are employed to detect scene objects, whose labels are inserted into the corresponding slots to produce captions [17]. Retrieval-based methods instead map both images and texts into a joint semantic space, search for the most similar image to a given query using a distance metric, and directly assign the retrieved image's caption to the query [18]. Although straightforward, both approaches suffer from limited flexibility and often yield descriptions with low diversity and adaptability.

In contrast, encoder–decoder-based methods attract greater attention for their ability to produce more fluent and diverse captions. Qu et al. [19] systematically compare combinations of pretrained convolutional neural networks (CNNs) and recurrent neural networks (RNNs), showing that using VGG-19 [23] as the image encoder and long short-term memory (LSTM [24]) as the text decoder achieves the best results. Liu et al. [20] extend this by integrating features from multiple ResNet50 [25] layers within the encoder to capture multiscale information, while the decoder incorporates self-attention and aggregation modules for more precise description generation. Further improvements come from Yang et al. [22], who introduce hierarchical feature aggregation with cross-modal alignment to enhance

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joint representations, and Meng et al. [21], who propose a global grouped attention mechanism to improve local feature modeling and contextual reasoning. More recently, the evolution of encoder–decoder architectures has given rise to vision–language models (VLMs), which leverage large-scale pretraining on massive image–text pairs to achieve stronger cross-modal alignment. For example, Wang et al. [26] proposed a single-stream encoder–decoder design for joint vision–language modeling, while Li et al. [27] introduced a dual-stream design connected through a bridging module. These architectures substantially enhance cross-modal interactions, leading to captions that are both more semantically accurate and more descriptively diverse.

Despite these advancements, all the above methods rely heavily on abundant labeled image–text pairs for training. In practical remote sensing scenarios, however, large quantities of unlabeled imagery are far more common, whereas paired image–text data remain scarce. This imbalance creates a significant obstacle for fully supervised approaches, limiting their applicability in real-world RSIC tasks.

B. Semi-Supervised Image Captioning

SSL improves model performance by combining limited labeled data with abundant unlabeled data, integrating supervised signals from labeled samples with the distributional characteristics of unlabeled ones. In image captioning, SSL shows strong potential to enhance generalization, reduce reliance on labeled data, and achieve competitive results in low-resource settings.

Early research studies in semi-supervised image captioning explore retrieval-based strategies. For example, Liu et al. [28] introduced a self-retrieval module that evaluates caption quality via image–text similarity, enabling the use of unlabeled data. Similarly, Kim et al. [29] proposed a pseudolabeling framework in which a discriminator verifies image–text pairs. These methods facilitate the use of unlabeled data but perform poorly when similar labeled samples are insufficient.

Beyond retrieval-based methods, self-training and adversarial learning approaches emerge as promising directions. Jain et al. [14] adapted the noisy student framework [13] to image captioning by introducing feature perturbations and label augmentation. Their method applies object dropout to perturb image features at the representation level and employs a caption paraphraser trained on unpaired text to refine pseudolabels, thereby mitigating erroneous predictions and diversifying generated captions. Yang et al. [11] propose cross-modal prediction and relation consistency (CPRC), which supervises caption generation from both informativeness and representativeness perspectives. Unlike conventional pseudolabeling, CPRC leverages prediction consistency—where image predictions act as soft labels—and relation consistency to align the relational distribution of generated captions with that of augmented images. In parallel, Kim et al. [10] introduced a semi-supervised framework that leverages large-scale unpaired images and captions alongside paired data, assigning pseudolabels through adversarial learning to capture the joint distribution of images and texts. Similarly, Liang et al. [12] employed a generative adversarial network to synthesize images from captions and used these generated pairs as additional training signals. In summary, these methods enhance pseudolabel quality and strengthen cross-modal alignment, but their reliance on confidence thresholds or adversarial heuristics still prevents effective use of informative low-confidence samples, underscoring the need for SSL frameworks that can integrate supervision across the entire confidence spectrum.

III. Methodology

Prior to delving into the framework (U²DP) proposed in this article, it is essential to define the problem setting and outline the key challenges that need to be addressed. Then, we provide an overview of our U²DP for semi-supervised RSIC, followed by a detailed explanation of its individual components. Finally, we formulate the objective function and provide the corresponding algorithmic process.

A. Problem Statement

Semi-supervised RSIC aims to generate accurate and semantically rich textual descriptions for remote sensing images by leveraging both limited labeled data and abundant unlabeled data. In practice, however, high-quality image–text pairs in the remote sensing domain remain scarce due to the high cost and expertise required for annotation. This scarcity poses a key challenge in fully exploiting large-scale unlabeled data to enhance model performance, motivating the development of approaches that can effectively integrate supervision signals from both labeled and unlabeled sources.

Formally, we define the semi-supervised RSIC dataset as: $\mathcal{D} = \{\{\mathbf{v}_i, \mathbf{w}_i\}_{i=1}^{N_l}, \{\mathbf{v}_j\}_{j=N_l+1}^{N_l+N_u}\}$, where $\mathbf{v} \in \mathbb{R}^{W\times H\times 3}$ denotes an image of dimensions $W \times H$ with three color channels, and $\mathbf{w}$ represents the corresponding caption. N_l and N_u ($N_l \ll N_u$) represent the number of labeled and unlabeled instances, respectively. Unlike traditional uni-modal tasks (e.g., target classification), image captioning seeks to learn a mapping function between images and text via a caption generator, thereby $\mathbf{w}$ serves as the supervision signal during the training phase. Next, we present the definition of the semi-supervised RSIC.

Definition 1 (Semi-Supervised RSIC): Given a limited set of labeled remote sensing image–caption pairs $\{\mathbf{v}_i, \mathbf{w}_i\}_{i=1}^{N_l}$ and abundant unlabeled data $\{\mathbf{v}_j\}_{j=N_l+1}^{N_l+N_u}$, our objective is to train a caption generator G that fully utilizes the unlabeled instances.

B. Framework

It is notable that our method, as a general semi-supervised framework, is compatible with any state-of-the-art (SOTA) captioning model. Given an image instance $\mathbf{v}$, the target of G is to generate a natural language sentence $\hat{\mathbf{w}}$ to describe the image. In this article, the base model G follows the widely-used encoder–decoder paradigm, and we consider two alternative architectures to ensure a fair comparison. One adopts a conventional CNN-based encoder and LSTM-based decoder framework (i.e., AoANet [30]), while the other utilizes a transformer-based model [21].

The framework of our U²DP is illustrated in Fig. 2. For illustrative purposes, we take AoANet as G. For each image

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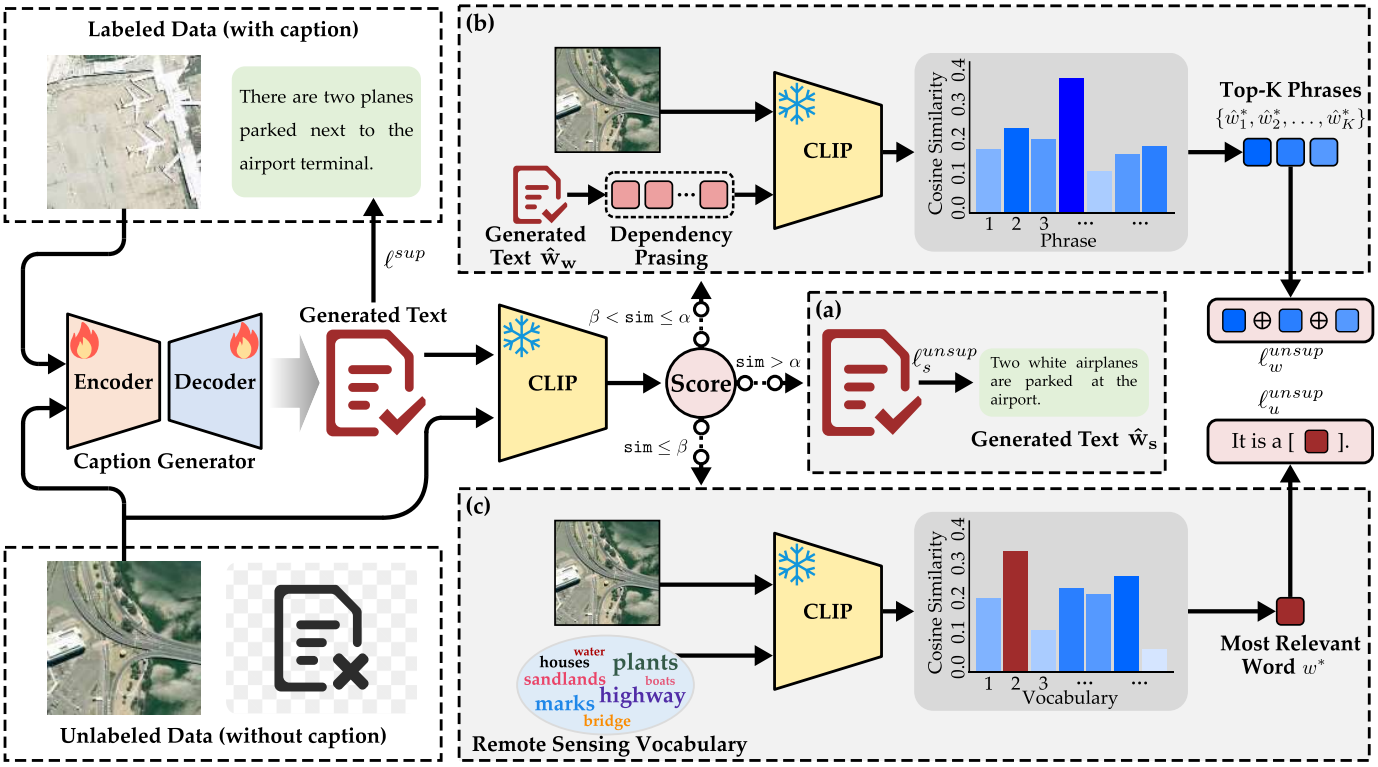


Fig. 2. Overall framework of U^{2}DP. Given an input image, the caption generator produces a textual description. For labeled data, a supervised cross-entropy loss ℓ^{sup} optimizes the alignment between the generated sentences and their corresponding annotations. For unlabeled data, the quality of the generated text is first evaluated by computing the semantic similarity between visual and textual embeddings in CLIP's joint embedding space. Then, a tri-level pseudosupervision strategy is constructed based on similarity score (i.e., (a) strongly matched caption, (b) weakly matched caption, and (c) unmatched caption).

$\mathbf{v}$ randomly sampled from the dataset $\mathcal{D}$, its visual embedding is first extracted by the encoder, implemented with faster R-CNN [31]. The refining module then applies an attention mechanism to dynamically refine the visual embedding for language generation. Finally, the decoder, implemented as an LSTM model [24], generates the final output sequence $\hat{\mathbf{w}}$. This process can be formulated as

$$\hat{\mathbf{w}} = \texttt{Decoder}\left(\texttt{Refining}\left(\texttt{Encoder}\left(\mathbf{v}\right)\right)\right). \tag{1}$$

Given this formulation, the model is trained with two primary objectives.

1) *Supervised Loss*, designed for labeled data (i.e., image–caption pairs), is computed using cross-entropy loss between the generated caption $\hat{\mathbf{w}}$ and its corresponding ground-truth caption $\mathbf{w}$.
2) *Unsupervised Loss*, designed for unlabeled data (i.e., unannotated images), is computed using cross-entropy loss between the generated caption $\hat{\mathbf{w}}$ and its corresponding pseudosupervision signal. Specifically, we measure the semantic similarity between the visual embedding of the input image and the textual embedding of the generated caption. Based on this similarity score, we define a tri-level pseudosupervision strategy to facilitate unsupervised learning.
   a) *Strongly Matched Caption:* Generated text with high similarity is directly used as the pseudosupervision signal.
   b) *Weakly Matched Caption:* For text with moderate similarity, key phrases are extracted and concatenated to form the pseudosupervision signal.
   c) *Unmatched Caption:* For low-similarity text, a salient object is predicted from the remote sensing vocabulary as a pseudolabel, which is then integrated into a template to form the pseudosupervision signal. Further details are provided in Section III-D.

C. Supervised Loss

The supervised loss is designed for labeled data and is computed using a generative cross-entropy loss, denoted as ℓ^{sup}. Given an image $\mathbf{v}$, the caption generator G produces a sentence sequence $\hat{\mathbf{w}}$ to describe the image. The supervised loss ℓ^{sup} is then calculated based on the generated sequence $\hat{\mathbf{w}}$ and the ground-truth caption $\mathbf{w} = \{w_1, w_2, \ldots, w_T\}$, where T is the sentence length

$$\ell^{\text{sup}} = -\sum_{t=1}^{T} \log p\left(w_t \mid w_{1:t-1}\right) \tag{2}$$

where $w_{1:t-1}$ denotes the ground-truth sequence up to the $(t-1)$th time step. The term $p(w_t \mid w_{1:t-1})$ represents the conditional probability of generating the word w_t given the preceding sequence $w_{1:t-1}$. The goal of ℓ^{sup} is to maximize the likelihood of the generated sentence $\hat{\mathbf{w}}$ matching the ground-truth caption $\mathbf{w}$.

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D. Unsupervised Loss

The unsupervised loss, denoted as ℓ^{unsup}, is designed to leverage unlabeled data by constructing adaptive pseudosupervision signals. To measure cross-modal semantic similarity, we employ CLIP [16], a VLM with an image encoder $f_v(\cdot)$ and a text encoder $f_t(\cdot)$, jointly pretrained on large-scale image–text pairs. For each unlabeled data $\mathbf{v}$, the caption generator G first produces a sentence sequence $\hat{\mathbf{w}}$. Then, we evaluate the quality of this generated caption by computing the semantic similarity $\texttt{sim}(\mathbf{v}, \hat{\mathbf{w}})$ between the visual embedding $f_v(\mathbf{v})$ and the textual embedding $f_t(\hat{\mathbf{w}})$ in CLIP's joint embedding space

$$\texttt{sim}(\mathbf{v}, \hat{\mathbf{w}}) = \frac{f_v(\mathbf{v})^\top f_t(\hat{\mathbf{w}})}{\|f_v(\mathbf{v})\|_2 \, \|f_t(\hat{\mathbf{w}})\|_2}. \tag{3}$$

Here, $\|\cdot\|_2$ represents the L_2 norm of the embedding. According to the similarity score, the generated captions are dynamically categorized into three quality levels: strongly matched ($\hat{\mathbf{w}}_\mathbf{s}$), weakly matched ($\hat{\mathbf{w}}_\mathbf{w}$), or unmatched ($\hat{\mathbf{w}}_\mathbf{u}$)

$$\hat{\mathbf{w}} \rightarrow \begin{cases} \hat{\mathbf{w}}_\mathbf{s}, & \text{if } \texttt{sim}(\mathbf{v}, \hat{\mathbf{w}}) > \alpha \\ \hat{\mathbf{w}}_\mathbf{w}, & \text{if } \beta < \texttt{sim}(\mathbf{v}, \hat{\mathbf{w}}) \leq \alpha \\ \hat{\mathbf{w}}_\mathbf{u}, & \text{if } \texttt{sim}(\mathbf{v}, \hat{\mathbf{w}}) \leq \beta. \end{cases} \tag{4}$$

Next, we construct pseudosupervision signals based on different quality levels of the generated captions as follows.

1) *Strongly Matched Caption:* Strongly matched caption, $\hat{\mathbf{w}}_\mathbf{s}$, exhibits high semantic similarity with the input image. Therefore, we directly employ it as the reliable pseudosupervision signal, and the corresponding unsupervised loss ℓ_s^{unsup} is computed as

$$\ell_s^{\text{unsup}} = -\sum_{t=1}^{T} \log p\left(\hat{w}_t \mid \hat{w}_{1:t-1}\right) \tag{5}$$

where $\hat{w}_{1:t-1}$ denotes the pseudosupervision sequence generated by G up to time step $(t-1)$.

2) *Weakly Matched Caption:* For weakly matched caption, $\hat{\mathbf{w}}_\mathbf{w}$, which exhibits partial semantic similarity with the input image, our goal is to retain informative content while suppressing irrelevant elements. To this end, we extract key phrases through relevance analysis. Specifically, we first perform syntactic dependence parsing to segment $\hat{\mathbf{w}}_\mathbf{w}$ into candidate phrases. Then, we compute the cosine similarity between each phrase $\hat{p}_i$ and the visual embedding. The Top-K most relevant phrases $\{\hat{p}_1^*, \hat{p}_2^*, \ldots, \hat{p}_K^*\}$ are concatenated to form the refined pseudosupervision signal $\hat{\mathbf{w}}^*$. The process can be formulated as follows:

$$\left\{\hat{p}_1^*, \hat{p}_2^*, \ldots, \hat{p}_K^*\right\} = \underset{\hat{p}_i \in \hat{\mathbf{w}}_\mathbf{w}}{\text{Top-K}}\left(\texttt{sim}(\mathbf{v}, \hat{p}_i)\right) \tag{6a}$$

$$\hat{\mathbf{w}}^* = \left(\hat{p}_1^* \oplus \hat{p}_2^* \oplus \cdots \oplus \hat{p}_K^*\right) \tag{6b}$$

$$\ell_w^{\text{unsup}} = -\sum_{t=1}^{T} \log p\left(\hat{w}_t^* \mid \hat{w}_{1:t-1}^*\right) \tag{6c}$$

where $\hat{w}_{1:t-1}^*$ denotes the pseudosupervision sequence up to time step $(t-1)$, and $\oplus$ represents the concatenation operation for connecting the Top-K phrases.

3) *Unmatched Caption:* For unmatched caption, $\hat{\mathbf{w}}_\mathbf{u}$, which exhibits significant semantic deviation from the input image, our aim is to prevent harmful noise propagation by replacing unreliable text with minimal, domain-relevant supervision. We achieve this by predicting a salient word, w_i^Φ, from the remote sensing vocabulary Φ based on its similarity to the visual embedding. The most relevant word token, w^*, is selected as a pseudolabel and incorporated into a prompt template as the pseudosupervision signal. We can formulate this process as follows:

$$w^* = \arg\max_{w_i^\Phi \in \Phi} \texttt{sim}\left(\mathbf{v}, w_i^\Phi\right) \tag{7a}$$

$$\mathbf{w}^* = \texttt{template}\left(w^*\right) \tag{7b}$$

$$\ell_u^{\text{unsup}} = -\sum_{t=1}^{T} \log p\left(w_t^* \mid w_{1:t-1}^*\right) \tag{7c}$$

where $w_{1:t-1}^*$ denotes the pseudosupervision sequence generated using CLIP's prompt template [`It is a` w^*] up to time step $(t-1)$.

This tri-level pseudosupervision strategy serves as a discrete confidence-based filtering mechanism, whose design adaptively suppresses potential noise while retaining informative content across quality levels.

E. Overall Function

In summary, we derive the loss functions for processing captions at three quality levels. When combined with the supervised loss, the final loss function is formulated as follows:

$$\mathcal{L} = \ell^{\text{sup}} + \lambda\left(\ell_s^{\text{unsup}} + \ell_w^{\text{unsup}} + \ell_u^{\text{unsup}}\right) \tag{8}$$

where λ balances the supervised and unsupervised losses. To mitigate the weak text generation of G in early training, we set $\lambda = 0$ for the first γ epochs to warm up ℓ^{sup}, and then activate the full loss. The overall U²DP is detailed in Algorithm 1.

F. Model Inference

After training, the learned caption generator G is used to generate captions for unseen images during inference. Each test image, $\mathbf{v}$, is first processed by the encoder to extract visual embeddings, which are then refined and passed to the decoder. Finally, the decoder generates the caption, $\hat{\mathbf{w}}$, word by word in an autoregressive manner.

IV. Experiments

This section presents a comprehensive evaluation of the proposed U²DP framework on multiple RSIC datasets. We begin with the datasets and evaluation metrics, followed by the experimental settings. We then report the experimental results with detailed analysis, complemented by an ablation study to verify the effectiveness of each key component. Next, we explore hyperparameter selection, provide qualitative analysis through visual examples, and investigate the influence of labeled and unlabeled data ratios. We also examine the framework's compatibility with other methods, assess its efficiency in terms of parameter count and inference speed, and finally discuss current limitations and future directions.

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Algorithm 1 Pseudocode of Our U^{2}DP Method

Input : Training set $\mathcal{D} = \{\{\mathbf{v}_i, \mathbf{w}_i\}_{i=1}^{N_l}, \{\mathbf{v}_j\}_{j=N_l+1}^{N_l+N_u}\}$.
Output: Learned parameters θ of caption generator G.
INIT Initialize parameters θ^0, maximum epochs E, warm-up period γ, scaling factor λ, hyper-parameters α, β.
/* Warm up model via supervised learning. */
for $t = 0$ **to** $\gamma - 1$ **do**
  Train model with labeled data $\{\mathbf{v}_i, \mathbf{w}_i\}_{i=1}^{N_l}$ and update parameters θ based on Equation 2.
end
/* Train model with $\mathcal{D}$ via semi-supervised learning. */
for $t = \gamma$ **to** $E - 1$ **do**
  Generate captions $\hat{\mathbf{w}}$ for unlabeled data $\mathbf{v}$ via G.
  Evaluate and categorize $\hat{\mathbf{w}}$ with Equation 3 and 4.
  /* Construct the pseudo-supervision signal based on semantic similarity $\mathtt{sim}(\mathbf{v}, \hat{\mathbf{w}})$. */
  if $\mathtt{sim}(\mathbf{v}, \hat{\mathbf{w}}) > \alpha$ **then**
    Construct the pseudo-supervision signal using $\hat{\mathbf{w}}_\mathbf{s}$.
  else if $\beta < \mathtt{sim}(\mathbf{v}, \hat{\mathbf{w}}) \leq \alpha$ **then**
    Extract Top-K phrases in $\hat{\mathbf{w}}_\mathbf{w}$ with Equation 6a.
    Construct the pseudo-supervision signal with Equation 6b.
  end
  else if $\mathtt{sim}(\mathbf{v}, \hat{\mathbf{w}}) \leq \beta$ **then**
    Predict the most relevant word w^* from the remote sensing vocabulary Φ with Equation 7a.
    Construct the pseudo-supervision signal with Equation 7b.
  end
  Train model with Equation 8 and update parameters θ.
  Update $t = t + 1$.
end

A. Datasets

Following prior works [21], [32], [33], we adopt the widely-used Sydney-Captions [19], UCM-Captions [34], and RSICD [34] datasets for evaluation.

1) *The Sydney-Captions dataset* contains 613 remote sensing images (500 × 500 pixels, 0.5-m resolution) across seven land cover categories, with 497/58/58 for training/validation/testing, respectively. Each image is paired with five human-annotated captions integrating visual and environmental contexts. The dataset is challenging due to its small size and the intricate details of urban landscapes in medium-resolution imagery.
2) *The UCM-Captions dataset* comprises 2100 images (256 × 256 pixels) across 21 categories, split into 1680/210/210 for training/validation/testing, respectively. Each image is annotated with five contextual captions describing visual elements and functional characteristics, making the dataset a benchmark for spatial–semantic understanding in remote sensing.
3) *The RSICD dataset*, the largest open-source captioning dataset, contains 10 921 images (224 × 224 pixels) across 30 categories, split into 8734/1094/1093 for training/validation/testing, respectively. Each image has five captions describing spatial relations and environmental semantics. Its large scale benefits the model training, while complex interclass variations pose evaluation challenges.

B. Evaluation Metrics

We employ five widely-used metrics to evaluate model performance: BLEU [35], METEOR [36], ROUGE-L [37], CIDEr [38], and S_m^*. These metrics can be calculated by the COCO caption evaluation tool.[1]

1) *BLEU* is a widely-used metric that evaluates similarity between generated and reference texts by measuring n-gram overlap from unigrams to longer phrases.
2) *METEOR* extends BLEU by considering exact matches, stemming, synonyms, and paraphrases. Leveraging external linguistic resources, it captures semantic similarities beyond surface word overlap.
3) *ROUGE-L* measures similarity using the longest common subsequence (LCS) between generated and reference texts, incorporating both precision and recall to reflect structural and content preservation.
4) *CIDEr* evaluates the consensus between generated and reference captions by computing TF-IDF weighted n-gram similarities. By emphasizing both frequent and distinctive expressions across multiple References, it captures relevance while promoting linguistic diversity.
5) S_m^* is the arithmetic mean of BLEU-4, METEOR, ROUGE-L, and CIDEr, providing a balanced and comprehensive measure of caption quality that reflects both linguistic accuracy and semantic depth.

C. Experimental Settings

To align with the SSL framework and account for dataset scale, we allocated 20%, 20%, and 10% of the Sydney-Captions, UCM-Captions, and RSICD datasets as labeled data, respectively, while using the remaining samples as unlabeled data. For all semi-supervised methods, both labeled and unlabeled data are accessible during training, whereas supervised methods are trained exclusively on the labeled data. To ensure fairness, we adopt two alternative architectures as the base model G: CNN+LSTM as in AoANet [30] and transformer as in MG-Transformer [21] for experiments. We denote these two variants as U^{2}DP and U^{2}DP+, respectively. We utilize the pretrained model CLIP-base[2] to evaluate caption quality. For weakly matched or unmatched captions, we generate the pseudosupervision signal via the prompt template [`It is a __.`]. As for the training process, the model is trained for 50 epochs with a batch size of 10 using the ADAM optimizer. The learning rate is initialized at 0.0002 and is multiplied by 0.8 every three epochs. The hyperparameters are set to α = 0.27/0.26/0.26, β = 0.24/0.24/0.24, and K = 3/2/3 for the Sydney, UCM, and RSICD datasets, respectively. The loss balance factor λ is set to 0.2, and the warm-up period γ is set to 5. All experiments are conducted on a single NVIDIA RTX 3090 GPU.

D. Comparison With Existing Baselines

We conduct a comprehensive comparison of U^{2}DP with two categories of SOTA baselines: supervised and semi-supervised

[1] https://github.com/tylin/coco-caption
[2] Pretrained weights for CLIP-base.

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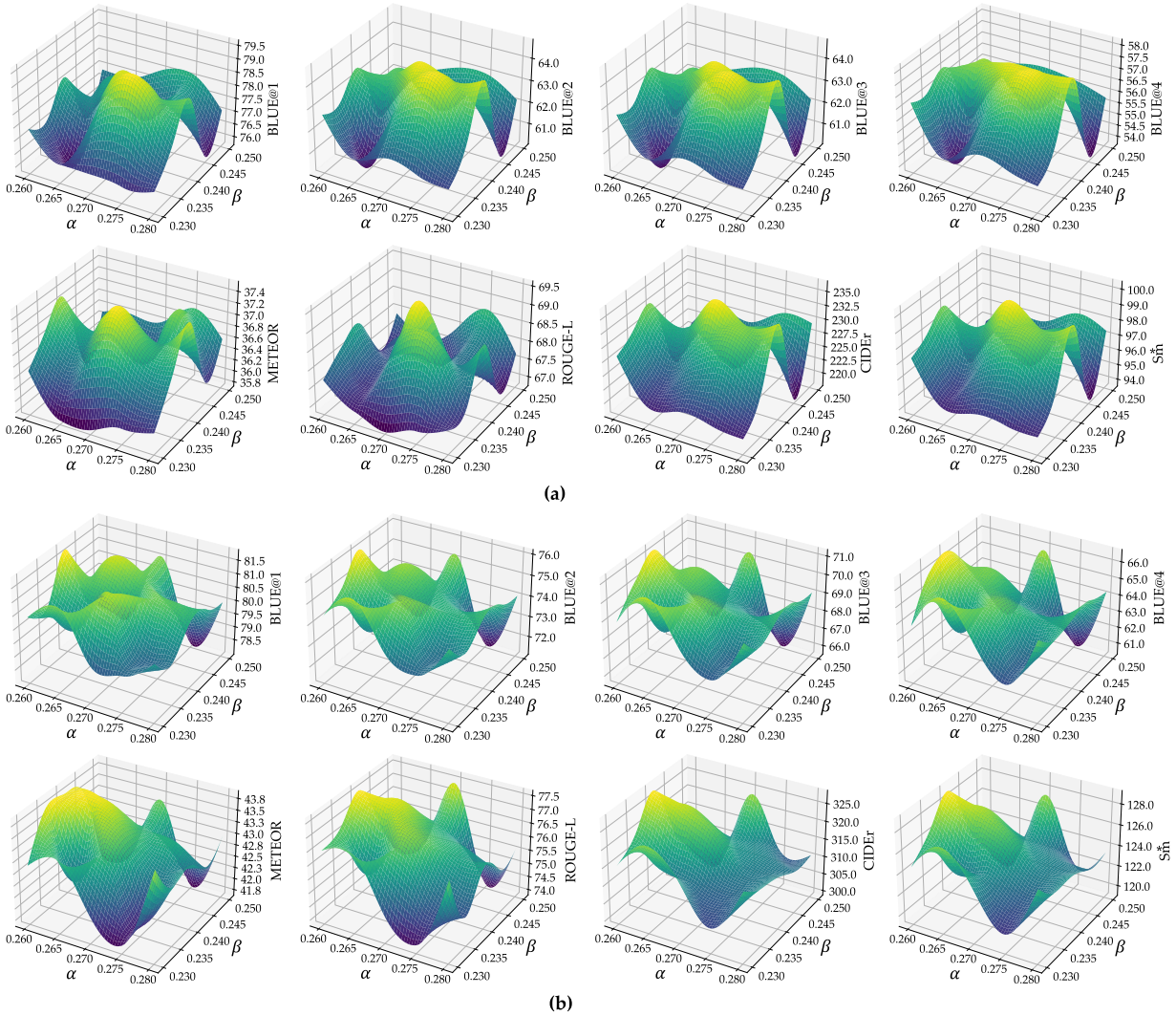


Fig. 3. Hyperparameter selection of α and β on (a) Sydney-Captions and (b) UCM-Captions datasets.

approaches. The former include AoANet [30], MLAT [20], HCNet [22], MG-Transformer [21], OFA [26], and BLIP-2 [27], while the latter consist of PPP [14], APLD [10], CPRC [11], and SIC-GAN [12]. Note that OFA and BLIP-2 are pretrained VLMs that cannot exploit unlabeled data during RSIC fine-tuning and are thus categorized as supervised approaches.

Tables I–III present the quantitative results of various SOTA baselines across multiple evaluation metrics on the Sydney-Caption, UCM-Caption, and RSICD datasets, respectively. It is important to note that supervised methods only make use of the labeled portion of the dataset while ignoring the unlabeled data. Referring to the results, we derive the following key observations.

1) *Advantage Over Supervised Methods:* We evaluate U^{2}DP against AoANet and HCNet, both based on conventional CNN+LSTM architectures, and U^{2}DP+ against MLAT and MG-Transformer, which leverage transformer architectures. In both cases, our methods consistently outperform their respective baselines. Specifically, on Sydney-Captions, U^{2}DP outperforms AoANet and HCNet in CIDEr by 7.5% and 34.9%, respectively, while U^{2}DP+ surpasses MLAT and MG-Transformer by 48.0% and 8.4%, respectively. These performance gains can be attributed to our semi-supervised framework, which allows the model to exploit information from the large proportion of unlabeled data. This approach improves the robustness and generalization capability of the model compared to purely supervised methods with limited labeled data.
2) *Advantage Over Pretrained VLMs:* We compare our method with two pretrained VLMs, OFA and BLIP-2. On UCM-Captions, U^{2}DP achieves competitive performance by effectively leveraging the remaining 80% unlabeled data. In contrast, despite their strong architectures, VLMs are trained in a purely supervised manner with only 20% labeled data and thus, cannot benefit from unlabeled samples. On the larger RSICD dataset, VLMs exhibit notable improvements as the amount of labeled data increases, outperforming U^{2}DP but still falling short of U^{2}DP+. This highlights that, when sufficient labeled data are available, model architecture is a critical factor in determining performance.

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TABLE I

QUANTITATIVE RESULTS WITH SOTA BASELINES ON THE SYDNEY-CAPTIONS DATASET. ALL RESULTS ARE REPORTED AS PERCENTAGES (%), WITH THE BEST PERFORMANCES HIGHLIGHTED IN BOLD

Methods	Encoder	Decoder	BLUE@1 ↑	BLEU@2 ↑	BLUE@3 ↑	BLUE@4 ↑	METEOR ↑	ROUGE-L ↑	CIDEr ↑	S_m^* ↑
AoANet [30]	Faster R-CNN	LSTM	77.1	68.0	60.5	53.5	37.3	69.2	228.4	97.1
MLAT [20]	ResNet-50	Transformer	75.8	69.5	62.6	56.0	36.8	67.9	209.0	92.4
HCNet [22]	ResNet-50	LSTM	73.9	65.8	58.7	52.8	38.2	69.2	201.0	90.3
MG-Transformer [21]	ResNet-152	Transformer	76.2	69.5	63.9	58.8	38.7	69.6	248.6	103.9
OFA [26]	Transformer	Transformer	75.2	66.5	59.9	53.9	37.9	68.5	226.3	96.7
BLIP-2 [27]	Transformer	OPT-LLM	75.7	66.6	59.1	52.7	37.9	70.1	238.5	99.8
PPP [14]	Faster R-CNN	LSTM	75.2	66.3	58.4	52.5	35.1	65.0	207.1	89.9
APLD [10]	Faster R-CNN	LSTM	75.9	67.1	58.7	52.6	34.9	65.9	216.7	92.5
CPRC [11]	Faster R-CNN	LSTM	77.9	69.4	62.9	54.7	37.2	69.2	233.9	98.8
SIC-GAN [12]	Transformer	Transformer	77.0	67.9	61.8	53.2	36.5	68.7	229.0	96.9
U^{2}DP	Faster R-CNN	LSTM	**78.8**	71.1	64.4	57.8	37.2	69.4	235.9	100.1
U^{2}DP+	ResNet-152	Transformer	77.1	**71.3**	**65.7**	**60.3**	**39.5**	**71.7**	**257.0**	**107.1**

TABLE II

QUANTITATIVE RESULTS WITH SOTA BASELINES ON THE UCM-CAPTIONS DATASET. ALL RESULTS ARE REPORTED AS PERCENTAGES (%), WITH THE BEST PERFORMANCES HIGHLIGHTED IN BOLD

Methods	Encoder	Decoder	BLUE@1 ↑	BLEU@2 ↑	BLUE@3 ↑	BLUE@4 ↑	METEOR ↑	ROUGE-L ↑	CIDEr ↑	S_m^* ↑
AoANet [30]	Faster R-CNN	LSTM	78.7	71.2	65.2	59.6	40.5	73.5	299.0	118.2
MLAT [20]	ResNet-50	Transformer	76.4	69.7	62.3	57.5	36.5	68.8	246.0	102.2
HCNet [22]	ResNet-50	LSTM	**83.3**	76.2	70.1	64.8	43.4	**79.2**	301.0	122.1
MG-Transformer [21]	ResNet-152	Transformer	81.3	75.9	71.2	67.1	43.9	75.7	332.0	129.7
OFA [26]	Transformer	Transformer	80.2	73.7	68.0	63.1	43.2	75.1	312.9	123.6
BLIP-2 [27]	Transformer	OPT-LLM	81.8	74.0	67.4	61.2	43.5	77.1	313.6	123.9
PPP [14]	Faster R-CNN	LSTM	78.2	71.8	67.1	62.2	42.0	75.4	305.4	120.5
APLD [10]	Faster R-CNN	LSTM	77.1	70.3	64.9	60.5	40.8	74.4	299.7	118.9
CPRC [11]	Faster R-CNN	LSTM	79.3	72.2	66.1	62.8	41.6	74.4	307.5	121.6
SIC-GAN [12]	Transformer	Transformer	78.4	70.7	65.5	61.3	39.8	73.4	298.3	118.2
U^{2}DP	Faster R-CNN	LSTM	80.1	73.8	68.7	64.0	42.8	75.6	314.6	124.3
U^{2}DP+	ResNet-152	Transformer	82.0	**76.3**	**71.7**	**67.5**	**44.8**	76.7	**352.5**	**135.4**

3) *Advantage Over Semi-Supervised Methods:* Our proposed U^{2}DP framework consistently outperforms semi-supervised baselines (PPP, APLD, CPRC, and SIC-GAN) across all metrics. Specifically, U^{2}DP achieves a 9.2% increase in CIDEr over PPP and a 16.3% increase over SIC-GAN on UCM-Captions. U^{2}DP+, which integrates a pretrained transformer backbone, achieves SOTA performance, underscoring the benefits of large-scale pretraining. The superiority over other semi-supervised methods mainly stems from our tri-level pseudosupervision mechanism, which adaptively suppresses noisy or ambiguous pseudolabels and reduces error propagation from unreliable supervision. This enables our model to generate more accurate and semantically rich captions, especially in scenarios with limited labeled data and abundant unlabeled data.
4) *Consistent Improvement Across Datasets:* The superior performance of our method is evident across all datasets, indicating its robustness and adaptability to diverse remote sensing scenarios. Particularly, on RSICD, which presents more complex scenes, our U^{2}DP+ still achieves significant improvements over all baselines, demonstrating its capability to generalize effectively across challenging datasets.

E. Ablation Study

In this section, we conduct extensive ablation experiments on two representative datasets (i.e., Sydney-Captions and UCM-Captions) to demonstrate the effectiveness of each component of our U^{2}DP. The experimental results are presented in Tables IV and V.

The experimental results demonstrate consistent trends across both datasets. Taking Table IV as an example, the baseline variant achieves a CIDEr score of 227.8% and an S_m^* score of 97.1%. To fully leverage the potential of unlabeled data, we progressively integrate three unsupervised loss components. The first component, ℓ_s^{unsup}, constructs pseudosupervision signals using strongly matched captions. By incorporating this component, we observe that the CIDEr and S_m^* scores increase by 0.8% and 0.6%, respectively. These results indicate that ℓ_s^{unsup} contributes to marginal performance improvements. When integrating the second component, ℓ_w^{unsup}, which handles weakly matched captions, the model achieves additional gains of 4.2% in CIDEr and 1.5% in S_m^*. This improvement suggests that extracting key phrases from weakly matched captions effectively reduces interference from irrelevant content, thereby strengthening the model's learned knowledge. Finally, we incorporate ℓ_u^{unsup}, leading

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TABLE III

QUANTITATIVE RESULTS WITH SOTA BASELINES ON THE RSICD DATASET. ALL RESULTS ARE REPORTED AS PERCENTAGES (%), WITH THE BEST PERFORMANCES HIGHLIGHTED IN BOLD

Methods	Encoder	Decoder	BLUE@1 ↑	BLEU@2 ↑	BLEU@3 ↑	BLUE@4 ↑	METEOR ↑	ROUGE-L ↑	CIDEr ↑	S_m^* ↑
AoANet [30]	Faster R-CNN	LSTM	60.1	43.2	32.9	25.9	23.8	44.5	71.5	41.4
MLAT [20]	ResNet-50	Transformer	60.9	43.7	33.4	26.6	23.6	45.2	72.8	42.1
HCNet [22]	ResNet-50	LSTM	60.2	42.4	32.5	25.9	23.0	43.3	73.4	41.4
MG-Transformer [21]	ResNet-152	Transformer	62.2	49.5	40.9	34.7	28.0	52.9	193.8	77.4
OFA [26]	Transformer	Transformer	62.8	47.2	40.7	33.6	26.5	53.1	192.2	76.4
BLIP-2 [27]	Transformer	OPT-LLM	**63.4**	48.8	41.8	34.1	27.3	53.8	196.9	78.0
PPP [14]	Faster R-CNN	LSTM	60.2	44.2	34.5	28.1	24.0	46.3	75.5	43.5
APLD [10]	Faster R-CNN	LSTM	59.1	42.7	33.5	25.8	23.1	44.9	72.2	41.5
CPRC [11]	Faster R-CNN	LSTM	60.4	43.1	32.8	25.9	23.7	44.6	71.7	41.5
SIC-GAN [12]	Transformer	Transformer	59.7	42.2	32.5	24.2	21.6	43.6	68.5	39.5
U²DP	Faster R-CNN	LSTM	60.1	44.4	34.7	28.1	24.2	46.4	76.2	43.7
U²DP+	ResNet-152	Transformer	**63.4**	**51.9**	**42.3**	**36.8**	**28.8**	**54.3**	**213.3**	**83.3**

TABLE IV

ABLATION STUDIES ON SYDNEY-CAPTIONS DATASET, WHERE B@N, *M*, *R*, AND *C* DENOTE BLEU@N, METEOR, ROUGE-L, AND CIDER, RESPECTIVELY. ALL RESULTS ARE REPORTED AS PERCENTAGES (%), WITH THE BEST PERFORMANCES HIGHLIGHTED IN BOLD

Methods			Sydney-Captions							
ℓ_s^{unsup}	ℓ_w^{unsup}	ℓ_u^{unsup}	B@1	B@2	B@3	B@4	M	R	C	S_m^*
✗	✗	✗	77.1	68.0	60.5	53.5	37.3	69.2	227.8	97.1
✓	✗	✗	78.4	70.4	63.2	56.1	37.3	**69.5**	228.6	97.7
✓	✓	✗	78.3	70.7	64.1	57.7	**37.4**	69.0	232.8	99.2
✓	✓	✓	**78.8**	**71.1**	**64.4**	**57.8**	37.2	69.4	**235.9**	**100.1**

TABLE V

ABLATION STUDIES ON UCM-CAPTIONS DATASET, WHERE B@N, *M*, *R*, AND *C* DENOTE BLEU@N, METEOR, ROUGE-L, AND CIDER, RESPECTIVELY. ALL RESULTS ARE REPORTED AS PERCENTAGES (%), WITH THE BEST PERFORMANCES HIGHLIGHTED IN BOLD

Methods			UCM-Captions							
ℓ_s^{unsup}	ℓ_w^{unsup}	ℓ_u^{unsup}	B@1	B@2	B@3	B@4	M	R	C	S_m^*
✗	✗	✗	78.7	71.2	65.2	59.6	40.5	73.5	299.0	118.2
✓	✗	✗	77.7	70.9	65.6	60.8	40.7	73.9	303.5	119.7
✓	✓	✗	79.8	72.5	66.6	61.2	41.8	74.8	311.3	122.3
✓	✓	✓	**80.1**	**73.8**	**68.7**	**64.0**	**42.8**	**75.6**	**314.6**	**124.3**

to a further improvement of 3.1% in CIDEr and 0.9% in S_m^*. This phenomenon demonstrates that introducing external knowledge for unmatched captions significantly contributes to comprehensive performance improvement.

The progressive integration of all components yields complementary gains, substantially boosting performance. This validates our tri-level pseudosupervision strategy for effectively exploiting unlabeled data while suppressing noise.

F. Hyperparameter Selection

In calibrating the proposed U²DP, we identify four hyperparameters: α and β in (4), which control the quality levels of generated captions; λ in (8), which balances the contributions of different losses; and K in (6a), which determines the number of key phrases extracted from the weakly matched caption.

1) Effectiveness of α and β: To investigate the impact of different α and β on the captioning process, we vary α within the range {0.260, 0.265, 0.270, 0.275, 0.280} and β within the range {0.230, 0.235, 0.240, 0.245, 0.250} on the Sydney-Captions and UCM-Captions datasets. Fig. 3 illustrates the model performance on the validation set, revealing the following observations: 1) a β value around 0.24 yields higher and more stable results across both datasets and 2) for α, the optimal value is approximately 0.27 for Sydney-Captions and 0.26 for UCM-Captions. This discrepancy can be attributed to the simpler sentence structures in UCM-Captions compared to Sydney-Captions, which lowers the threshold for strongly matched data and reflects the inherent differences in caption content between the two datasets. Moreover, these results indicate that α and β jointly control the trade-off between noise suppression and information retention, underscoring the importance of fine-tuning them to fully exploit the benefits of unlabeled data.

2) Effectiveness of λ: To explore how the unsupervised loss influences the model's performance during training, we systematically evaluate the balance factor λ across values

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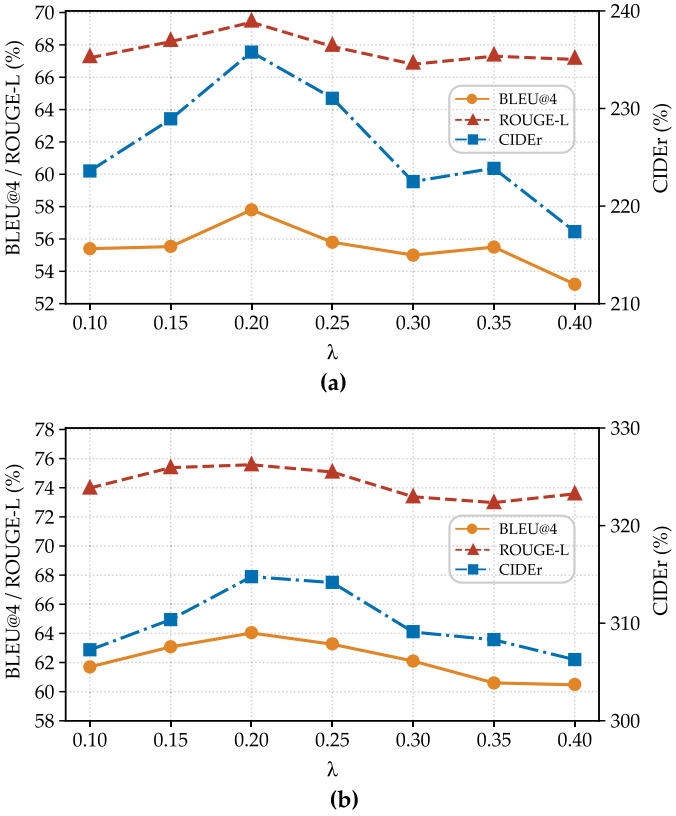


Fig. 4. Hyperparameter selection of λ on (a) Sydney-Captions and (b) UCM-Captions datasets.

{0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4} on both the Sydney-Captions and UCM-Captions datasets. Fig. 4 presents the metric trajectories, revealing two significant findings.

1) As λ increases, the performance of U^{2}DP initially improves and then declines. This indicates that while a moderate incorporation of unsupervised loss enhances learning, an excessive reliance on unlabeled data gradually weakens the model's ability to effectively leverage labeled data, ultimately leading to performance degradation.
2) U^{2}DP achieves optimal performance when λ is set to 0.2 on both datasets, suggesting a well-balanced trade-off between supervised and unsupervised loss components.

3) Effectiveness of K: We further examine the effect of extracting different numbers of phrases from weakly matched captions on model performance. The experimental results, as presented in Table VI, reveal dataset-specific optimal configurations: for the Sydney-Captions dataset, extracting three phrases that are most semantically relevant to the input image yields maximum performance improvement, while for the UCM-Captions dataset, optimal performance is achieved with two extracted phrases. This divergence in optimal phrase numbers directly corroborates our previous findings regarding the informative differences in caption content between the two datasets. Specifically, the richer content in Sydney-Captions allows for more effective utilization of multiple phrases to construct the pseudosupervision signal, whereas the relatively simpler captions in UCM-Captions benefit from a more conservative approach to phrase extraction.

TABLE VI

PERFORMANCE OF U^{2}DP WITH THE NUMBER OF PHRASES EXTRACTED FROM WEAKLY MATCHED CAPTION, WHERE B@N, *M*, *R*, AND *C* DENOTE BLEU@N, METEOR, ROUGE-L, AND CIDER, RESPECTIVELY. ALL RESULTS ARE REPORTED AS PERCENTAGES (%)

Methods	Sydney-Captions							
	B@1	B@2	B@3	B@4	M	R	C	S_m^*
K = 1	78.4	71.1	64.4	57.8	36.7	68.7	233.1	99.1
K = 2	78.7	**71.9**	**65.0**	**58.4**	37.0	68.5	235.1	99.8
K = 3	**78.8**	71.7	64.4	57.8	**37.2**	**69.4**	**235.9**	**100.1**
K = 4	77.7	69.9	63.3	56.8	36.4	68.0	225.8	96.8
K = 5	78.0	70.4	63.7	57.0	36.9	69.0	225.5	97.1
Methods	UCM-Captions							
	B@1	B@2	B@3	B@4	M	R	C	S_m^*
K = 1	79.6	73.4	68.4	63.6	42.7	74.8	**315.1**	124.1
K = 2	**80.1**	**73.8**	**68.7**	**64.0**	**42.8**	**75.6**	314.6	**124.3**
K = 3	80.0	73.4	68.6	62.4	42.7	75.5	311.9	123.4
K = 4	79.6	73.5	67.9	61.8	42.6	74.7	311.7	123.1
K = 5	79.4	72.5	66.8	61.2	42.4	74.2	304.3	120.5

G. Qualitative Analysis and Attention Visualization

1) Qualitative Analysis: To intuitively demonstrate the effectiveness of our method, we present qualitative analysis between AoANet, PPP, the proposed U^{2}DP, and human-labeled ground-truth captions in Fig. 5. The results show that U^{2}DP generates significantly more accurate and comprehensive descriptions than AoANet and PPP, highlighting its superior ability to capture both high-level scene semantics and fine-grained details.

For example, in the third subfigure of Fig. 5(a), AoANet and PPP incorrectly classify the scene as a "residential area," whereas U^{2}DP correctly identifies it as a "meadow," demonstrating its enhanced scene recognition capability. Moreover, U^{2}DP excels at producing detailed descriptions that closely align with image content. As illustrated in the first subfigure of Fig. 5(c), AoANet and PPP generate captions that mention only "green trees," "viaduct," and "buildings," whereas U^{2}DP successfully incorporates additional elements, such as "cars," highlighting its ability to capture finer details often missed by existing methods. This comparison underscores U^{2}DP's strength in generating comprehensive and contextually rich descriptions, particularly in recognizing smaller objects and complex scene compositions.

2) Attention Visualization: To provide a more intuitive understanding of our method, we visualize the attention maps generated by U^{2}DP across different training stages. As shown in Fig. 6, the remote sensing image contains a "tennis court," a "stadium," several "buildings," and surrounding "plants." In the early stage of training, the model focuses mainly on "plants," misclassifying large areas as "forest" and overlooking key objects such as the "tennis court." This is reflected by a low CLIP score of 0.238, leading the caption to be categorized as Unmatched. As training progresses, the model gradually shifts attention toward the "buildings," indicating a partial improvement in semantic alignment. By the 42nd epoch, the model attends to most key objects, focusing on the "stadium" and "tennis court," resulting in a strongly matched caption with a higher CLIP score.

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AoANet: A river with many white buildings.
PPP: There are some white buildings on the industrial area.
Ours: An industrial area with many white buildings and some roads.
GT1: A river goes through a industrial area with many white buildings.
GT2: There are many white buildings on the banks of a river.



AoANet : There are some green bushes and white bunkers on the meadow.
PPP: A residential area with houses arranged neatly and some roads go through this area.
Ours: Some white buildings on the meadow while some roads go through this area.
GT1: A meadow with some green bushes on it while a highway passed by.
GT2: There is a meadow with scattered bushes and some houses on it.



AoANet: A residential area with houses arranged neatly while some roads go through this area.
PPP: A residential area with houses arranged and some green bushes on it .
Ours: A meadow with some green bushes and some bunkers.
GT1: A meadow with some green bushes on it while a highway passed by.
GT2: There is a meadow with scattered bushes and some houses on it.



AoANet: This is a part of deep green sparkling sea with a runway beside.
PPP: This is a part of deep green sparkling sea.
Ours: There are some marking lines on it while some lawns beside.
GT1: A narrow runway with some lawns beside.
GT2: A narrow rectangular runway goes through the lawn.

(a). Sydney-Captions



AoANet: A storage tank with grey roofs is in the lawn.
PPP: Two storage tanks are on the lawn.
Ours : A storage tank is on the lawn with some plants and a road beside.
GT1: A storage tank is on the ground with some houses and a road beside.
GT2: There is one white storage tank on the ground with some houses and a road beside.



AoANet : There is a small baseball diamond.
PPP: Lots of airplanes are stopped atthe airport.
Ours : There are many airplanes on the ground.
GT1: There are many airplanes atthe airport.
GT2: Many white airplanes are stopped neatly atthe airport.



AoANet : A river with grey roofs is in the sparse residential area.
PPP: A villa is in the sparse residential area.
Ours : A house with plants surrounded is in the sparse residential area.
GT1: A house with is surrounded by bushes and sands in the sparse residential area.
GT2: This is a sparse residential area with a house surrounded by bushes and sands.



AoANet : This is a beach with dense forest.
PPP: Lots of plants flourish on both banks of the river.
Ours : Waves slapping a white sand beach.
GT1: This is a beach with blue sea and white sands.
GT2: Waves beat on the beach over and over again.

(b). UCM-Captions



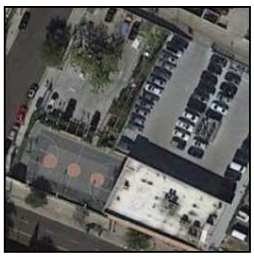
AoANet : Many green trees are near a viaduct.
PPP: Some buildings and green trees are near a viaduct.
Ours : Many buildings and green trees are near a viaduct with many cars.
GT1: Many viaducts are built in this area.
GT2: There are many tall trees on both sides of the road.



AoANet : Many boats are in a port near several green trees.
PPP: Many boats are in a port.
Ours : Many boats are in a port with a grey roof beside.
GT1: A large number of white ships parked in the port.
GT2: The harbour is surrounded by a grey roof.



AoANet: It is a piece of green meadow.
PPP: It is a piece of khaki bareland.
Ours : It is a piece of yellow bareland.
GT1: A bare trail on the wide land.
GT2: No plants were planted on the bare land.



AoANet : Many boats are in a port near several buildings and green trees.
PPP: Many buildings and green trees are in a commercial area.
Ours : A basketball field and some trees near the parking lot.
GT1: A parking lot and a basketball field are near two roads.
GT2: A basketball field is nextto a small parking lot.

(c). RSICD

Fig. 5. Examples of captions generated by AoANet, PPP, our proposed U²DP, and the corresponding ground-truth captions. Detailed and accurate words are marked in green, while words inconsistent with the image content are marked in red. (a) Sydney-Captions. (b) UCM-Captions. (c) RSICD.

These observations demonstrate the effectiveness of our tri-level pseudosupervision strategy in progressively improving pseudolabel quality. This strategy can be regarded as a confidence-based filtering mechanism that not only mitigates the adverse impact of noisy or ambiguous captions but also adaptively refines pseudolabel reliability across different confidence levels, thereby enabling the model to concentrate on semantically relevant entities and further strengthen image–text alignment.

H. Influence of the Labeled and Unlabeled Data

To systematically investigate the impact of data composition on model performance, we conduct two sets of controlled

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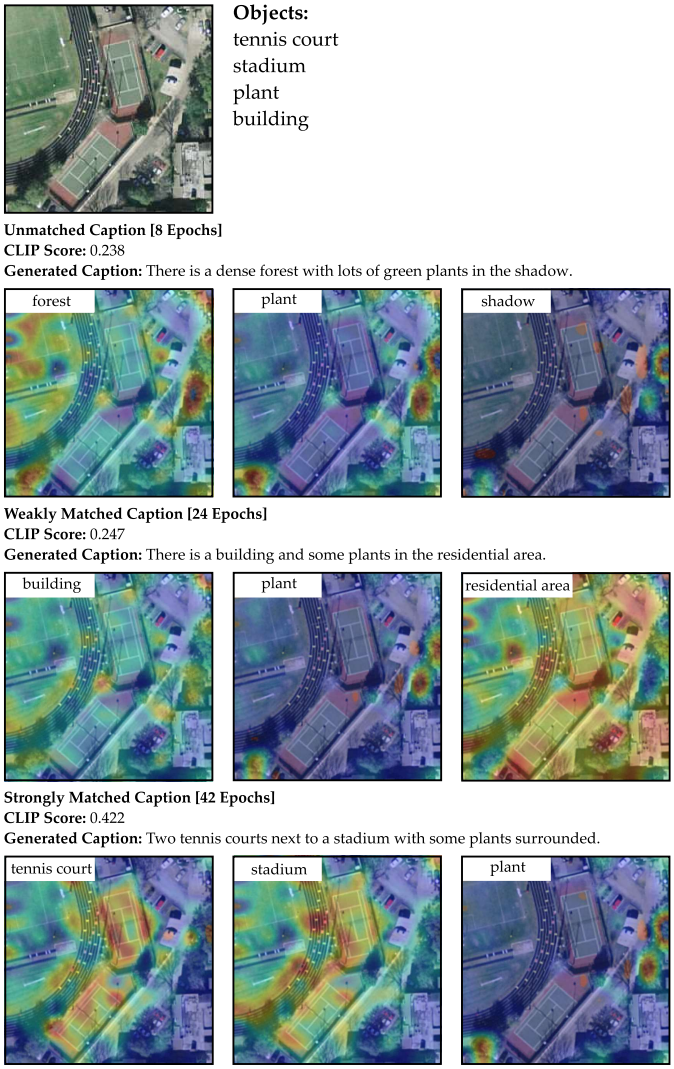


Fig. 6. Comparison of heatmaps across different training stages. Attention intensity is indicated by color, with red denoting higher attention and blue lower attention.

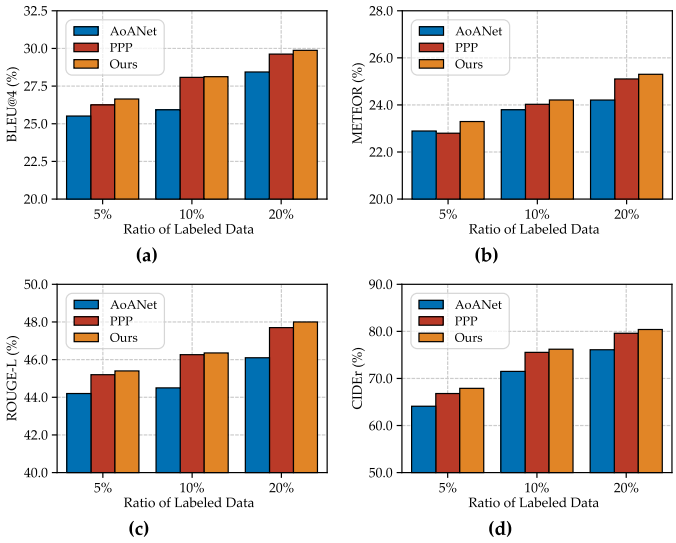


Fig. 7. Relationship between caption performance with different ratios of labeled data on the RSICD dataset. (a) BLEU@4. (b) METEOR. (c) ROUGE-L. (d) CIDEr.

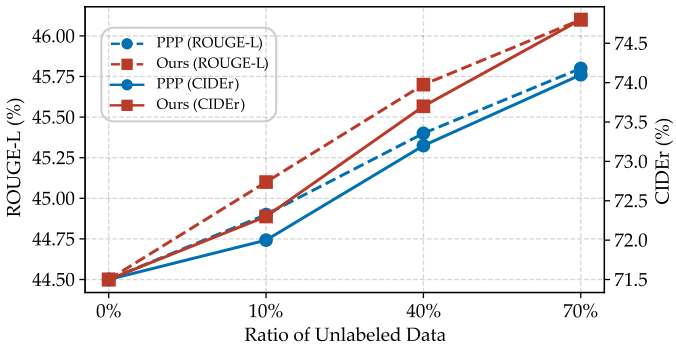


Fig. 8. Relationship between captioning performance with different ratios of unlabeled data on the RSICD dataset.

experiments focusing on labeled and unlabeled data, respectively. We first examine the influence of labeled data by progressively increasing its ratio while monitoring performance metrics. As illustrated in Fig. 7, U^{2}DP demonstrates superior learning efficiency compared to AoANet and PPP, leading to significantly faster performance improvement as the ratio of labeled data increases. This observation suggests that our framework effectively leverages labeled data to enhance the generator's learning capability.

Furthermore, we examine the impact of unlabeled data by fixing the ratio of labeled data at 1% while varying the ratio of unlabeled data across {10%, 40%, 70%}. The results, presented in Fig. 8 (focusing on two critical metrics, ROUGE-L and CIDEr), reveal a consistent performance improvement as the ratio of unlabeled data increases. This finding confirms that U^{2}DP effectively leverages unlabeled data to enhance model training by constructing pseudosupervised signals.

I. Compatibility With Other Methods

Our U^{2}DP framework is designed to maximize the utilization of unlabeled data by constructing diverse pseudosupervised signals. This design enables U^{2}DP to function as a model-agnostic plugin that can be seamlessly integrated with various existing methods. As demonstrated in Fig. 9, U^{2}DP consistently enhances the performance of different supervised approaches. Experimental results demonstrate significant improvements across key evaluation metrics, including BLEU@4, METEOR, ROUGE-L, and CIDEr, when integrated with other models. These findings highlight U^{2}DP's strong compatibility with diverse architectures and its potential as a universal enhancement module for the semi-supervised RSIC task.

J. Model Efficiency

We briefly analyze the efficiency of our model in comparison with other SOTA methods. As reported in Table II, U^{2}DP+ achieves the highest captioning performance on the UCM-Captions dataset. At the same time, Fig. 10 illustrates that it maintains a competitive parameter count and inference speed. It is worth noting that both CPRC and U^{2}DP are built upon the AoANet architecture, leading to similar parameter counts and inference times, with the main difference arising from the adopted semi-supervised training strategies.

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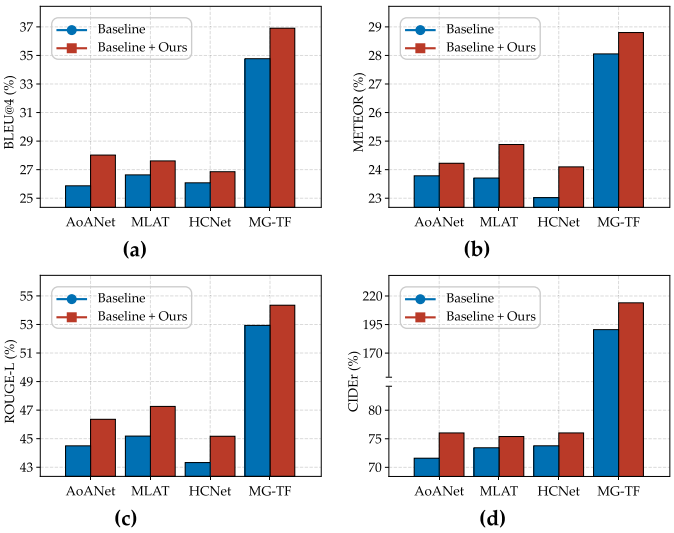


Fig. 9. Robust performance of U²DP across various supervised methods on the RSICD dataset. (a) BLEU@4. (b) METEOR. (c) ROUGE-L. (d) CIDEr.

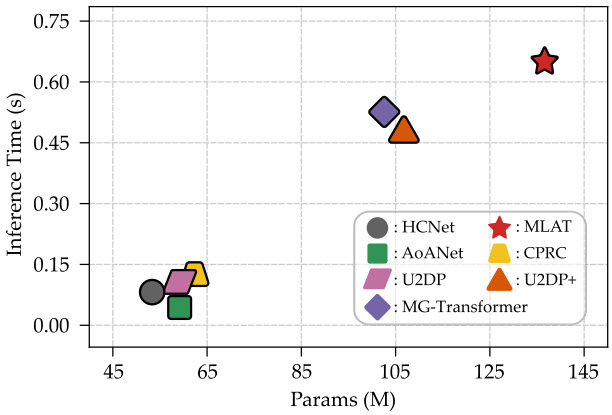


Fig. 10. Comparison of model size and inference time across different methods on the UCM-Captions dataset.

K. Limitations

Although U²DP achieves strong results on multiple RSIC datasets, several limitations remain. First, as with most semi-supervised approaches, its performance can be affected by the distribution and diversity of the unlabeled data. When the unlabeled samples are not sufficiently representative, the potential benefits tend to diminish. Second, while using a remote sensing-specific vocabulary improves pseudolabel quality in unmatched cases, adaptively expanding or updating this vocabulary could further enhance adaptability to novel scenarios. Third, the current framework has been evaluated mainly on captioning tasks. Extending it to other remote sensing tasks, such as cross-modal retrieval or broader multimodal understanding, remains an open direction. Addressing these aspects could further improve the robustness and generalization of our method.

V. Conclusion

In this article, we presented U²DP, a novel semi-supervised RSIC framework that fully exploits unlabeled data through adaptive supervision reconstruction. Unlike existing methods that rely on selective filtering, U²DP categorizes unlabeled data into three levels (strongly matched, weakly matched, and unmatched) based on their semantic similarity with generated captions, and then constructs tailored pseudosupervision signals by integrating model-generated predictions and the pseudolabel retrieved from the domain-specific vocabulary. This approach enables effective learning without discarding data, thereby enhancing caption quality. Extensive experiments on three widely used RSIC datasets, namely Sydney-Captions, UCM-Captions, and RSICD, demonstrate that our method consistently surpasses SOTA baselines, achieving notable improvements in CIDEr scores. Ablation studies further confirm the complementary benefits of the hierarchical pseudosupervision strategy and the effectiveness of CLIP-based semantic alignment in improving caption generation.

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